

APS106

more OOP!

Week 11 | Lecture 1 (11.1)

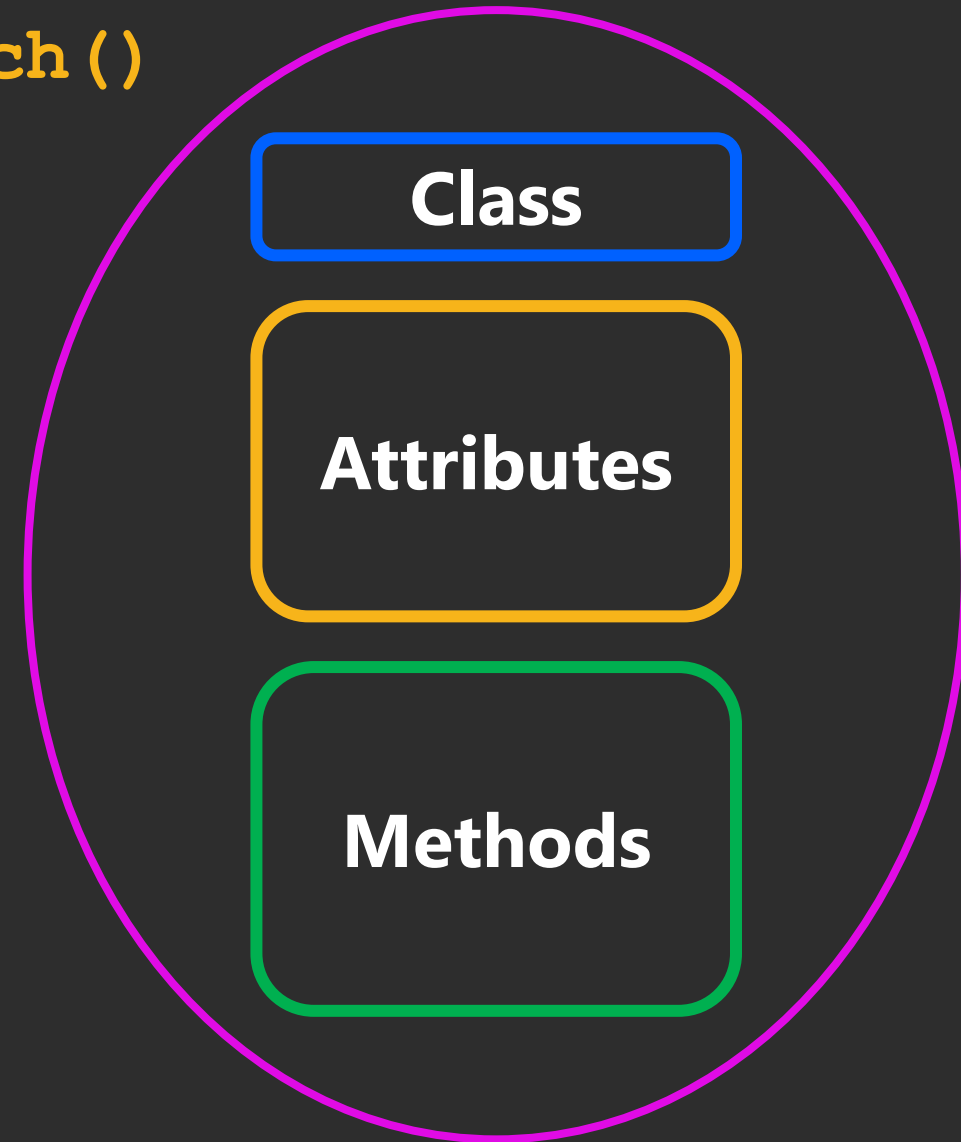
if nothing else, write **#cleancode.**

Encapsulation.

```
seb.forward(50)  
kitty.scratch()
```

Encapsulation

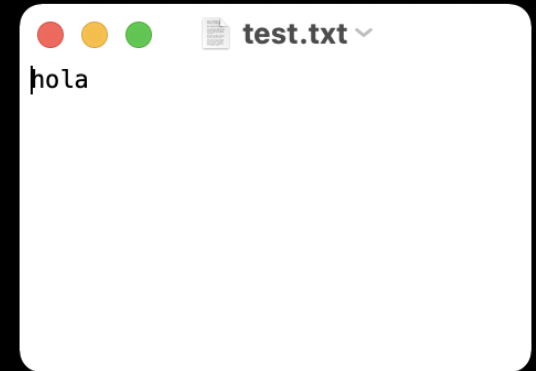
- The core of object-oriented programming is the organization of the program by **encapsulating** related **data** and **functions** together in an object.
- To encapsulate something means to enclose it in some kind of container.
- In programming, encapsulation means keeping **data** and the **code** that uses it in one place and hiding the details of exactly how they work together.



Encapsulate it.

- For example, each instance of class `file` keeps track of which file on the disk it is reading/writing and where it currently is on that file.
- The class hides the details of how it is done, so we (as programmers) can use it without needing to know how it is implemented.

```
f = open('test.txt', 'w')  
f.write('hola')  
f.close()
```

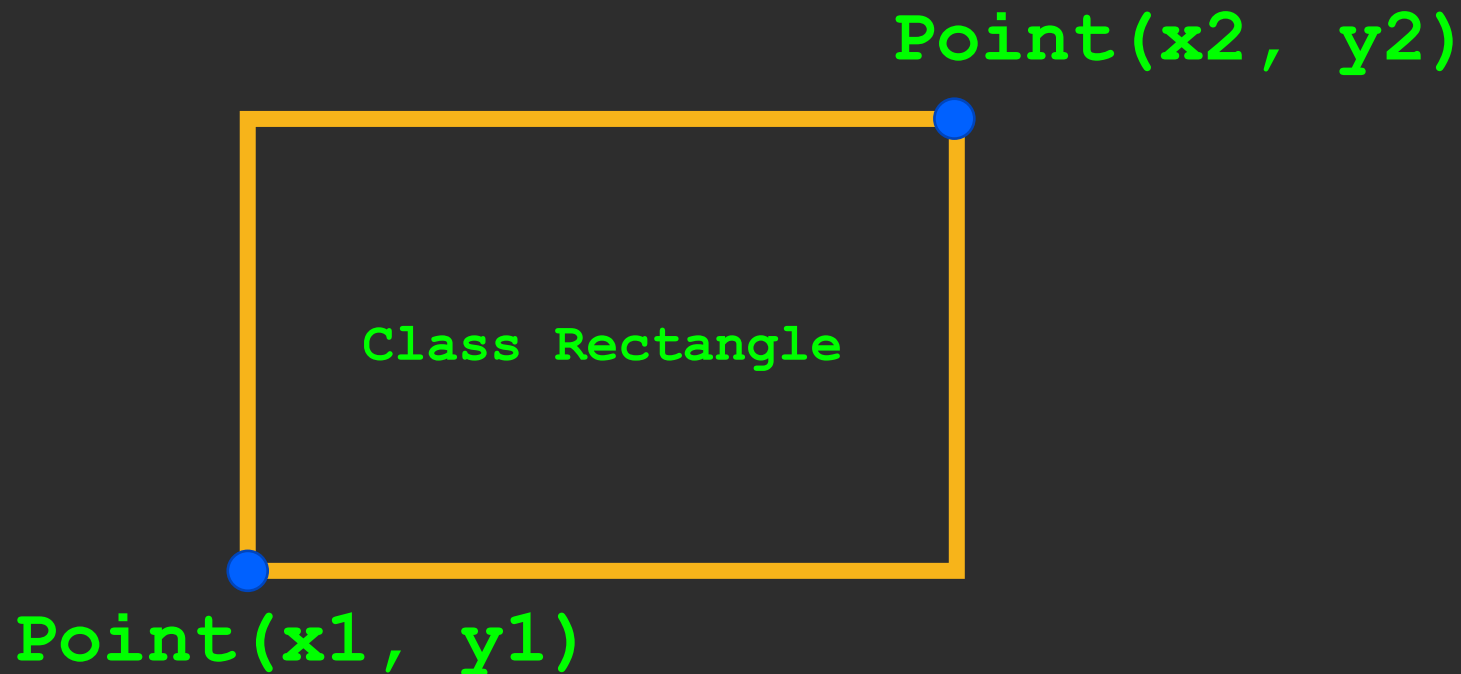


```
print(f)
```

```
<_io.TextIOWrapper name='test.txt' mode='w' encoding='UTF-8'>
```

Let's Code!

- Let's look at how this works in Python!
 - Point Class Recap.
 - Rectangle Class.



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Click Link:
1. Encapsulation

Printing Information.

- It would be nice to not have to write a print statement when we want to display attribute information:

```
>>> p = Point(3,4)
>>> print(p)
<__main__.Point object at 0x7fd8100778b0>
```

```
>>> p = Point(3,4)
>>> print(p.x, p.y)
3 4
```

- Is there a better way to encapsulate this? How about a method?

```
>>> p = Point(3,4)
>>> p.to_string()
'(3,4) '
>>> r = p.halfway(Point(5,12))
>>> r.to_string()
'(4,8) '
```

Printing Information.

- If we had a method, we just need to format the attributes into a string:

```
class Point:

    """A class that represents and manipulates 2D points"""

    def to_string(self):
        """
        (self) -> None
        Prints the (x, y) coordinate for the point.
        """
        print('(' + str(self.x) + ', ' + str(self.y) + ')')
```

- We could make a similar method for any class.

```
>>> p = Point(3,4)
>>> p.to_string()
'(3,4)'
>>> r = p.halfway(Point(5,12))
>>> r.to_string()
'(4,8)'
```

Printing Information.

- If every class we made had a **to_string()** method, great!
 - Easy to remember.
 - All formatting specific to printing an object would be encapsulated.
 - But they won't all have one...
- What if other people made their own method for printing and called it something else?
 - Then it would no longer be obvious how to quickly display information.
 - Is it called **to_string**? **tostring**? **2stringz**? **print_point**? **plz_print_me**?
- **Python has a clever trick for this!**



Printing Information.

Special Methods Magic Methods

These are part of Python's special object protocol.

- If we call our new method `__str__` instead of `to_string`, Python will use our code whenever it needs to convert Point to a string.

```
class Point:
    ...
def __init__(self, x=0, y=0):
    ...

len(x) → x.__len__()
print(x) or str(x) → x.__str__()
bool(x) → x.__bool__()
```

```
def __str__(self):
    return '(' + str(self.x) + ',' + str(self.y) + ')'
```

```
>>> p = Point(3,4)
>>> print(p)
'(3,4)'
```

That's better!

When we call `print(obj)`, then `obj.__str__()` is called to find out what string to print.

Let's Code!

- Let's look at how this works in Python!
 - `__str__` method.

switching from procedural programming
to object oriented programming be like

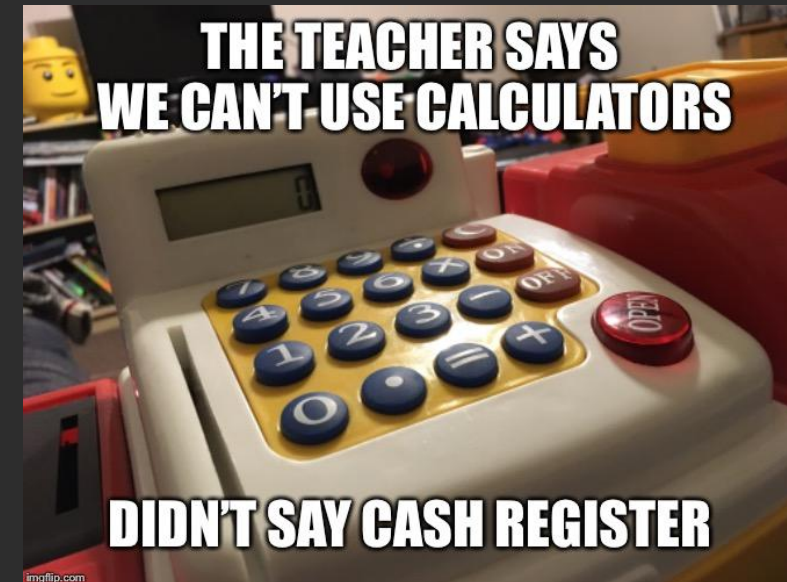


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Click Link:
2. Printing Objects

Let's Build a Cash Register!

- We want to build a cash register program.
What features do we want?
 - Track number of:
 - Loonies.
 - Toonies.
 - \$5 bills.
 - \$10 bills.
 - \$20 bills.
 - Accepts cash.
 - Removes cash.
 - Calculates value of contents.
 - Print the entire contents.



Breakout Session!

- Let's look at how this works in Python!
 - Building a cash register:
 - Attributes.
 - Methods.
 - Including `__str__` method.



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Click Link:
**3. Building a Cash
Register**

Let's Code!

- We are going to model a solar system using OOP.
 - Two classes: Planet and SolarSystem.
 - **Planet**: name, distance, radius, color, angle.
 - **SolarSystem**: add/remove planets, visualize, animate.
 - Custom module handles the visualization.
 - Kepler's Third Law for orbital periods.
- Let's look at how this works in Python!
 - **Planet** class.
 - **SolarSystem** class.
 - Visualization and animation.

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Click Link:
4. Solar System

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